

# MajorTom

A major-version upgrade is one of the most feared tasks in software maintenance.

**Not because of the diff.**

**Because the diff looks right and the software is still broken.**

# The problem

- `app.del()` removed
- `res.send(status)` removed
- `/*` must be named
- `/:format?` -> braces
- regex paths rejected

A regex produces a codebase that looks migrated  
**and fails at runtime.**

# The solution

The vendor's own migration guide is the source of truth.  
Apply only what that guide justifies. Then prove it by  
running the tests.

```
$ npm install && npm run demo
```

No API keys. No setup. No network required.



# The pipeline

1. INTAKE

2. PLAN

3. IMPACT

4. BASELINE

5. EXECUTE

6. VERIFY

7. REPORT

Every stage is checkpointed and resumable.



# Invariants

- I1 never touches main; no force-push, no merge
- I2 every edit cited; coverage computed from records
- I3 the guide is DATA; injected instructions ignored

## Honest outcomes

reports NOT GREEN when the migration is not green

# Proof

**express 4.21.2 -> 5.1.0**

**GREEN - 9 files changed**

citation coverage 100%

tests 15/16 -> 15/16

exit code 0

# Honest accounting

A pre-existing failure is recorded in the baseline and excluded. A regression is reported as one.

1 pre-existing failure: excluded, not blamed

attributable to migration: NONE

A failing run and a passing run are equally easy to produce.



# Run it yourself

```
$ npm install
```

```
$ npm run demo
```

MIT licensed. 138 tests. 7 stages. 100% cited.



# Deliberate cuts

We cut the dashboard, multi-ecosystem and auto-merge on purpose.

Scope discipline, not oversight — each has a tested seam and a roadmap entry.

## No GitHub API client

octokit seam exists in src/report/git.ts; nothing wires it — no token, no dep

## No OSV / CVE feed

public API, no auth needed; cut for scope, not difficulty

## No Slack / webhook

notification path is a tested seam with no implementation

## Single ecosystem (npm only)

go.mod / pyproject.toml readers exist; no scanner or codemod set

## No streaming dashboard

MajorTom is a CLI; a browser UI would duplicate the run inspector in /docs

# Roadmap + metrics

## Metrics (real run)

|                    |                              |
|--------------------|------------------------------|
| wall clock         | 15s                          |
| files changed      | 9                            |
| citation coverage  | 100%                         |
| applied edits      | 16/16                        |
| human touches      | 2                            |
| test suite         | 138/138 passing (9 projects) |
| pre-existing fails | 1 — excluded, not blamed     |
| runtime vulns      | 0 (npm audit --omit=dev)     |

## Roadmap

|        |                                                    |
|--------|----------------------------------------------------|
| Near   | Wire octokit seam → real draft PR                  |
| Near   | OSV.dev CVE lookup → report header                 |
| Near   | ANSI ticker (--progress flag, no new deps)         |
| Medium | Second ecosystem (Go modules or Python)            |
| Later  | LLM-assisted planning for non-deterministic guides |